

Day 01 – Tokenizers & SLM

Name: Moturi Lalitha Sowjanya

Team: 4

Week: 02

Date: 17-06-26

1) OBJECTIVE

To study the concept of Small Language Models (SLMs), understand the role of tokenizers in language models, explore a suitable application domain for an SLM, identify the problem it can solve, and understand its usefulness and potential users.

2) RESEARCH TOPICS

2.1. Tokenizers

2.1.1. What is a tokenizer?

A tokenizer is a tool that breaks text into smaller units called tokens. These tokens can be words, sub words, characters, or sentences. Language models cannot understand raw text directly, so tokenizers convert text into tokens before it is processed by the model.

2.1.2. Why is tokenization needed?

Tokenization is needed because language models cannot understand raw text directly. They can only process numbers. A tokenizer breaks text into smaller units called tokens and converts them into numerical IDs that the model can understand.

It is needed because it:

- text into a format the model can process.
 - Helps the model understand the meaning and structure of the text.
 - Makes text processing faster and more efficient.
 - Helps the model handle unknown or complex words by breaking them into smaller parts.
-

2.1.3 Types of tokenizers

1. Word Tokenizer

- Splits text into individual words.

- **Example:**

"I love AI"
↓
["I", "love", "AI"]

2. Sub word Tokenizer

- Splits uncommon words into smaller meaningful parts.
- **Example:**

"unhappiness"
↓
["un", "happi", "ness"]

3. Character Tokenizer

- Splits text into individual characters.
- **Example:**

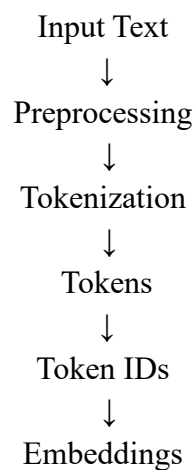
"AI"
↓
["A", "I"]

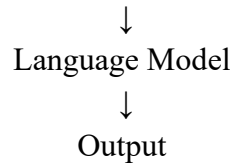
4. Sentence Tokenizer

- Splits text into sentences.
- **Example:**

"Hello. How are you?"
↓
["Hello.", "How are you?"]

2.1.4. Workflow of Tokenization





2.1.4. How Tokenization Impacts Model Performance

- Helps the model understand text more accurately.
 - Reduces the size of the vocabulary by using subwords.
 - Improves the handling of unknown or rare words.
 - Makes training and text processing more efficient.
 - Better tokenization generally leads to better model performance.
-

3) PROPOSED SLM FOR THE SELECTED DOMAIN

3.1. Selected Domain

Healthcare

3.2. Proposed SLM

Medical Information Assistant

3.3. Problem Statement

People often struggle to understand medical information due to complex terminology and the large amount of information available from different sources. The proposed Medical Information Assistant SLM aims to provide clear, reliable, and easy-to-understand information about diseases, medicines, symptoms, and treatments using trusted healthcare resources.

3.4. Why is the Proposed SLM Useful in the Healthcare Domain?

- The proposed Medical Information Assistant SLM is useful in the healthcare domain because it provides quick, reliable, and easy-to-understand medical information from trusted healthcare resources.
- It helps users understand diseases, symptoms, medicines, treatments, and preventive care without having to search through multiple sources.
- It can also support healthcare professionals and medical students by providing relevant information efficiently, making access to healthcare knowledge faster and more convenient.

3.5. Target Users

The proposed Medical Information Assistant SLM can be used by:

- **Patients** – To understand diseases, symptoms, medicines, and treatments in simple language.
- **Healthcare Professionals** – To quickly access medical information and reference materials.
- **Medical Students** – To support learning and understand medical concepts.
- **Caregivers** – To learn about patient care, medications, and health conditions.
- **General Public** – To access reliable health information and increase health awareness.